

PathScribe by ForMedrix AI Admin Guide

System Administration & Configuration

For Administrators and Lead Pathologists

1. Administration Overview

Administrators access PathScribe by ForMedrix AI configuration from the Home screen. All configuration is under the **Configuration** tile, divided into four tabs: **System**, **Staff**, **Protocols**, and (when HL7/FHIR is active) **LIS Integration**.

Important — Roles and access: The Admin role provides configuration access only. Admins cannot see clinical case data by design.
A user who needs both clinical and admin access must hold both the Pathologist and Admin roles simultaneously.

PathScribe operates in two deployment modes. Your deployment mode is set at onboarding and determines which features are active:

Mode	Description	Typical customer
Co-Pilot	PathScribe provides AI assistance and synoptic reporting alongside your Cerner Epic, Beigle, Sunquest, etc.	Hospitals, long-term care, etc.
Standalone LIS	PathScribe is the system of record — case management, test ordering, reports, etc. — and full Cerner, etc. integration.	Independent pathology groups, reference labs, etc.

2. Staff Management

PathScribe distinguishes two categories of people — **Staff** (internal users who log in) and **Physicians** (external ordering doctors who never log in). Understanding this distinction is essential before setup.

Category	Who they are	Log in?	Managed in
Staff	Pathologists, Residents, Administrators	Yes — login	Configuration → Staff
Physicians	External ordering / referring doctors	Never — direct	Configuration → System → Physicians

2.1 Adding a New Staff Member

- Navigate to **Configuration** → **Staff** and click **Add User**.
- Complete all required fields (see table below). Click **Save**.
- The new user can log in immediately with their email address.

Field	Required	Notes
First & Last Name	Yes	Appears on reports, audit logs, and in-app display.
Email	Yes	Used for login (if SSO not configured) and notifications.
Role(s)	Yes	One or more: Pathologist, Resident, Admin. Permissions are the union of all assigned roles.
NPI Number	No	Required for billing-related workflows.
License Number	No	State/jurisdiction medical licence — jurisdiction-specific.
Department	No	e.g. Surgical Pathology, Hematopathology, Neuropathology.
Phone	No	Used for directory and urgent notification routing.
Voice Profile	Yes	EN-US or EN-GB — drives voice command recognition accuracy.

Paediatric Access	No	Enable to allow this user to see paediatric cases. Disabled by default.
Status	Yes	Active / Inactive. Inactive users cannot log in.

2.2 Roles and Permissions

Roles are not mutually exclusive. A staff member can hold multiple roles; their permissions are the union of all assigned roles.

Role	Case worklist	Sign-out	Configuration	Notes
Pathologist	Full access	Primary sign-out	None	Core clinical role. Full case lifecycle.
Resident	Full access	Co-sign only	None	Sign-out gating controlled by usePermission config.
Admin	None	None	Full access	Cannot access clinical case data by design.

2.3 Deactivating and Reactivating Users

Click a user row to expand their record, then click **Deactivate**. The account is preserved for audit purposes but login is blocked. Click **Reactivate** to restore access.

2.4 Paediatric Access Control

Cases involving patients below your institution's configured paediatric age threshold are only visible to users with **Paediatric Access** enabled. Users without this flag see the case row in the worklist with all PHI redacted. Enable or disable per user in their staff record.

3. Physician Directory

Navigate to **Configuration** → **System** → **Physicians** to manage the external ordering physician directory. Physicians appear on reports and drive result delivery routing. They do not log in and hold no permissions.

Field	Required	Notes
First & Last Name	Yes	Used on reports and result notifications.
NPI Number	Recommended	Used for billing validation and deduplication.
Fax / Email	At least one	Required for result delivery. Without this, reports cannot be sent to the physician.
Preferred Contact	No	Drives automatic routing: Email, Fax, or Portal.
Client Affiliations	No	Links physician to one or more submitting facilities.
Status	Yes	Active / Inactive / Unverified. Unverified physicians do not receive notifications.

Auto-created physicians: When an incoming case references an NPI not yet in the directory, PathScribe creates an Unverified physician record automatically.

Unverified physicians will not receive result notifications until an admin adds a contact method and sets the record to Active.

Check the Physicians table regularly for the amber Unverified indicator.

4. Role Management

Navigate to **Configuration** → **Staff** → **Roles** to manage role definitions and permissions.

4.1 Built-in Roles

Role	Case access	Config access	Can be deleted	Description
Pathologist	Yes	No	No	Full clinical case access and sign-out authority.
Resident	Yes	No	No	Case access with co-sign only. Sign-out gated by attending counter-
Admin	No	Yes	No	System configuration. No clinical access.
Physician	No	No	No	External directory only. No app access.

4.2 Custom Roles

- Click **Add Role**. Enter a name, description, and set Case Access and Configuration Access toggles.
- Custom roles can be fully edited and deleted. Built-in roles can only have their permissions adjusted.

4.3 Permissions

Each role has a granular permissions list. Click the **Permissions** tab on a role record to view and edit individual action permissions (e.g. "Can finalise reports", "Can delegate cases", "Can view AI suggestions", "Can amend finalized reports"). Permissions are enforced throughout the application.

5. Flag Management

Navigate to **Configuration** → **System** → **Flags**. Flags are visual markers on cases and specimens that communicate clinical context or required actions.

5.1 Case Flags vs Specimen Flags

- **Case flags** apply to an entire case (e.g. "Tumor Board — Thu 14:00", "QI Review Required", "Teaching Case").
- **Specimen flags** apply to a specific specimen within a case (e.g. "ER/PR/HER2 Pending", "BRAF Mutation", "Molecular Panel"). Specimen flags typically indicate pending or resulted ancillary tests.

5.2 Creating a Flag

- Click **Add Flag**. Enter a name and description.
- Select a colour (Red, Amber, Yellow, Blue, Green) and severity level (1 = informational, 5 = critical).
- Set the scope: Case flag or Specimen flag.
- For specimen flags, enter the **LIS Code** — the code used by your LIS to trigger this flag automatically on result receipt.
- Click **Save**. The flag is immediately available in the worklist and on cases.

5.3 Computational Flags

Computational flags represent tests ordered via the Computational Sidecar. To configure a computational flag, select the flag and expand the **Computational** section:

Data Source Type	Description	Required fields
LIS Endpoint	Direct REST call from PathScribe to the LIS	Secret ID, API Endpoint URL.
AI Extraction	AI proxy parses unstructured LIS text to extract results	Extraction endpoint URL.
PathScribe Entry	Pathologist enters the result directly in the Synoptic form	Synoptic form endpoint required.

Link one or more published synoptic protocols to a computational flag. When a pathologist orders that test, PathScribe suggests these protocols for addition to the case.

5.4 Delegation Types

Navigate to **Configuration** → **System** → **Delegation Types** to manage the types of case delegation available to pathologists.

- Click **+ New Type**. Enter a Label — the ID is auto-derived (e.g. "Peer Review" → PEER_REVIEW).
- Fill in Description, optional CPT Hint, accent colour, and Active toggle.
- Click **Create**. The new type immediately appears in the Delegate Modal for all active users.

System types (built-in) can only be toggled active/inactive. **Custom types** can be fully edited and deleted.

6. AI Configuration

Navigate to **Configuration** → **AI Settings**.

Setting	Default	Description
AI Provider	Anthropic	Select Anthropic Claude or Google Gemini as the active AI provider.
Model	claude-sonnet-4-20250514	Select specific model version. Anthropic Claude Sonnet 4 is recommended for clinical accuracy.
API Mode	Proxy	Proxy routes calls through the backend (production). Direct is development only.
Confidence Threshold	75%	Fields with AI confidence below this threshold show "AI: not found" instead of a suggestion.
Gross-Driven AI	On	AI analyses the gross description and pre-populates synoptic fields automatically.
Microscopic-Driven AI	On	AI re-analyses after microscopic text is entered, refining existing suggestions.
Auto-insert suggestions	Off	If on, AI suggestions are applied to synoptic fields without requiring pathologist confirmation.
Voice AI	Gemini active	Enable Gemini-powered voice command recognition.

Confidence Threshold: The threshold directly controls which fields appear in the AI Review triage modal before finalisation. Setting it lower means more fields pass through automatically; higher means more fields require explicit review before sign-out.

Important: Changes to AI settings take effect for new sessions. Active users must log out and back in to see updated AI behaviour.

7. System Configuration

Navigate to **Configuration** → **System**. The left sidebar gives access to all system-wide settings.

7.1 LIS Integration

Setting	Default	Description
LIS Integration Enabled	Off	Master switch. Enable only when the HL7/FHIR endpoint is configured and tested.
LIS Endpoint URL	(empty)	The HL7 or FHIR endpoint for PathScribe's outbound synoptic delivery.
LIS Owns Case Statuses	On	Case status is read-only in PathScribe — the LIS is authoritative. Turn off only in Standard mode.
Allow Post-Final Actions	On	Allows addenda and amendments after sign-out. Disable if your LIS controls this workflow.

7.2 Jurisdiction

Set the jurisdiction to match your institution's regulatory environment. This controls which SNOMED national release is used and which ICD-10 variant is active.

Jurisdiction	SNOMED release	ICD variant
US	SNOMED CT US Edition	ICD-10-CM
CA (Canada)	SNOMED CT CA Edition	ICD-10-CA

GB_EW (England & Wales)	SNOMED CT UK Edition	ICD-10 (WHO)
GB_SCT (Scotland)	SNOMED CT UK Edition	ICD-10 (WHO)
IE (Ireland)	SNOMED CT IE Edition	ICD-10 (WHO)

7.3 Identifier Formats

Configure the regex patterns used by the smart identifier box in Case Search to detect accession numbers and MRNs from typed input.

- Go to **Configuration** → **System** → **Identifier Formats**.
- Enter the regex pattern for your accession number format. The **Example** field lets you test the pattern live — a green ✓ confirms it matches.
- Enter the regex pattern for your MRN format and test similarly.
- Click **Save**. Detection updates immediately for all users.

Common accession formats:

S26-4200 → `^[A-Z]\d{2}-\d{4,6}$`

PATH-2026-04200 → `^PATH-\d{4}-\d{5}$`

12345678 (numeric only) → `^\d{6,10}$`

7.4 Specimens

The specimen dictionary drives autocomplete in the Case Search filter and in case creation. Add, edit, or deactivate specimen types here. Each specimen type has a name, category, and optional synonyms.

7.5 Subspecialties

Define subspecialty pools used in the delegation workflow. Subspecialties group pathologists for pool case assignment (e.g. Neuropathology Pool, Breast Pool). Cases can be routed to a subspecialty pool rather than an individual pathologist.

7.6 Terminology Services

Navigate to **Configuration** → **System** → **Terminology Services** to check the health of all connected terminology endpoints:

- SNOMED CT — national release endpoint.
- ICD-10 / ICD-11 / ICD-O — coding system endpoints.
- CPT — AMA CPT code endpoint (requires AMA licence — see implementation checklist).
- LOINC — LOINC endpoint (used for laboratory result mapping in the Computational Sidecar).

Each service shows a status badge (Connected / Degraded / Offline) and last-checked timestamp. A green badge is required before go-live.

8. Report Template Builder

The Report Template Builder allows administrators to create, edit, and manage the synoptic report templates used by pathologists. Templates are composed of reusable **Parts** assembled in a defined layout.

Key architecture: A **Part** is an independently-versioned module (header, body section, footer). A **Template** is an ordered assembly of Parts.
Updating a Part instantly updates every template that references it — no need to edit templates individually.
PathScribe ships with 13 pre-built CAP-compliant Parts. Review these before building from scratch.

8.1 Accessing the Template Builder

Navigate to **Configuration** → **Template Builder**. The builder has two tabs: **Part Library** and **Templates**.

8.2 Creating a Part

- Click **New Part** in the Part Library tab.
- Enter a name, description, part type (Header / Body / Footer), specialty, and optional subspecialty.
- Set the Standard (CAP, RCPATH, or Custom). Click **Create** to open the Part Builder canvas.

8.3 The Canvas and Palette

The canvas is a 12-column drag-and-drop editor. Drag components from the left palette onto the canvas. Components are displayed as cards with a coloured left accent strip.

Component	Description
Section	Collapsible group of related fields. Use for logical clinical groupings (Tumour Characteristics, Margins, Lymph Nodes).
Text Field	Single-line field bound to a case data key. Renders as "Label: Value" in the report.
Data Paragraph	Multi-line text slot filled from case data (e.g. gross description, microscopic description). Rich text supported.
Rich Text Block	Static text identical in every report — disclaimers, institutional statements, fixed clinical text.
Dropdown	Selection field with predefined options. Supports single and multi-select. Options defined in Inspector.
Number + Unit	Numeric input with unit label (e.g. 18 mm). Configurable min/max and decimal places.
Date	Date field. Configurable format. Can default to today.
Computed	Calculated field whose value is derived from an expression combining other field values.
Static Label	Heading or label at any typographic level (H1, H2, H3, body, caption).
Repeat Group	Container that iterates over a collection (e.g. specimens). Children rendered once per item.
Column Layout	Arranges children in 2, 3, or 4 flowing newsprint-style columns.
If Block	Conditional container — children shown only when a condition is true (e.g. "show only if HER2 positive").
Page Break	Forces a page break in the printed report at that point.
Header	Institutional letterhead. Configure elements shown (logo, accession, patient name). Scope to page 1 or all pages.

Footer	Page footer with page numbers and confidentiality notice. Scope to page 1 or continuation pages.
Image Embed	Embeds an image from a URL or case data binding key.

8.4 Canvas Interactions

- **Add:** Drag from Palette to canvas. Drop between existing cards to insert at a specific position.
- **Reorder:** Drag any card up or down within its container.
- **Resize columns:** Drag the right edge of a card to change its column span (1–12). Two cards at 6 columns each sit side by side.
- **Nest:** Drag a component into a Section, Repeat Group, If Block, or Column Layout to nest it. Containers show a disclosure triangle.
- **Delete:** Hover over a card and click the red trash icon that appears.
- **Select:** Click any card to load its settings in the Inspector panel (right side).

8.5 The Inspector Panel

The Inspector panel shows all configurable properties for the selected component:

- **General:** Label, Width (column span), Required toggle, Hide if Empty toggle, Page Break Before.
- **Component-specific:**
 - Text Field: Binding key, placeholder, validation regex, max length.
 - Dropdown: Binding key, options list (value|Label format), multi-select toggle, allow free text.
 - Number: Binding key, unit, decimal places, min/max range.
 - Section: Print heading, collapsible toggle, AI generation config (instruction, max tokens, temperature).
 - Column Layout: Number of columns (2, 3, or 4), column gap in pixels.
 - If Block: Condition expression builder (field, operator, value — supports AND/OR logic).
 - Header: Logo toggle, accession number toggle, patient name toggle, scope (page 1 / pages 2+ / all).
 - Footer: Page number format string, scope.
- **Label Formatting:** Position (adjacent / above / none), text transform, bold/underline, font size. Live preview shown.
- **Visibility Condition:** Show/hide rule applied to any component. Renders only when the condition is true.

8.6 AI Generation Configuration

Sections can be configured to allow AI-generated narrative content. Select a Section and scroll to **AI Generation** in the Inspector:

- Toggle **Enable AI generation** on.
- Enter a **System instruction:** be specific (e.g. "Write a two-sentence clinical summary of the tumour grade and lymph node status based on the synoptic data provided.").
- Set **Max tokens** (controls response length) and **Temperature** (0.0 = deterministic, 1.0 = varied; 0.3 recommended for clinical accuracy).

8.7 Previewing a Part

Click **Preview** in the canvas toolbar to open the Part Preview panel. This renders the Part using mock clinical data. Verify: labels, data bindings, conditional blocks, column layouts, page breaks, header/footer scoping.

Use the page size and margin controls to check across different print formats.

8.8 Assembling a Template

In the **Templates** tab, click **New Template**. Give the template a name, specialty, and standard.

The Template Assembly page has two panels:

- **Left panel — Part Library:** All available Parts filtered by specialty. Search, filter by type, and drag Parts into the assembly area.
- **Right panel — Assembly Canvas:** The assembled layout, divided into role zones:
 - **Page 1 Header** — header shown on the first page only.
 - **Pages 2+ Header** — continuation header for pages after the first.
 - **Body** — one or more body Parts stacked in sequence. Reorder by dragging within the body zone.
 - **Page 1 Footer** — footer for the first page.
 - **Pages 2+ Footer** — footer for continuation pages.

Minimum valid template: At least one Body Part is required to publish. Headers and footers are strongly recommended.

A template without a Page 1 Header shows a validation warning but can still be published.

8.9 Publishing a Template

Click **Publish** when the assembly is complete and the preview is correct. The template becomes available for selection in the synoptic reporting workflow. The version number increments automatically.

To update a published template: make changes and publish again. The new version is available immediately. In-progress cases retain the version that was active when they were started.

9. Action Registry & Voice Control

PathScribe has 139 registered actions — every action available via mouse or keyboard is also available by voice command. The Action Registry is at **Configuration** → **Action Registry**.

9.1 Architecture

The voice pipeline has four layers:

Layer	Function
Voice Provider	Captures microphone input via the Web Speech API. Normalises transcripts. Runs on every page.
Action Registry	Matches transcripts against registered voice triggers using exact match, then learned triggers, then fuzzy score.
Event Dispatcher	Fires a PATHSCRIBE_<ACTION_ID> custom DOM event. Handlers in AppShell and page components respond.
Learning Layer	Records unrecognised phrases. When a user confirms what they meant, the mapping is stored permanently.

9.2 Coverage

Coverage by application area:

Area	Actions
System, Navigation, Messages (AppShell)	49 actions
Worklist	30 actions
Synoptic Report and sub-components	60 actions
Search	3 actions
Configuration tab navigation	2 actions
Total	139 actions / 60 with keyboard shortcuts

9.3 Editing an Action

- Navigate to **Configuration** → **Action Registry**.
- Find the action by name or ID. Click to expand.
- Edit the **keyboard shortcut** or **voice triggers** (comma-separated phrases). Multiple triggers allow natural language variants.
- Click **Save**. Changes are stored in localStorage and applied immediately — no restart required.

Persistence: Administrator edits to shortcuts and voice triggers persist permanently across deployments. New actions added by developers are merged in automatically — existing customisations are never overwritten. To reset to defaults, clear the ps_action_registry key from localStorage.

9.4 Voice Context Awareness

The same phrase means different things on different screens. "Confirm" in the AI Review modal accepts a finding; "confirm" in the synoptic panel confirms a field value. The system knows the difference because handlers are registered per-page and per-modal, not globally.

10. Audit Log

Navigate to **Home** → **System Audit**. The audit log records all significant system events — immutable and append-only.

- User login and logout events.
- Case creation, status changes, finalisation, and amendments.
- AI suggestion accept and reject events (confidence scores recorded).
- User account create, edit, and deactivate.
- Configuration changes (all settings, flag, role, template edits).
- Delegation and pool claim events.
- Computational test orders, cancellations, and result receipt.

Each record shows the timestamp, user ID, user name, action, and the affected entity (case ID, user ID, etc.). Filter by date range, user, or event type. Click **Export CSV** for compliance reporting.

11. System Information

Click your initials badge → **System Information** to view:

- **Application:** product name, company, version, build date, environment (development / production).
- **AI Provider:** active provider, model name, API mode, Voice AI status.
- **API Connectivity:** live status of Anthropic and Gemini API connections.
- **Browser & System:** browser name and version, OS, screen resolution — for support diagnostics.

Click **Copy to Clipboard** to generate a formatted support report. The copy redacts the user's name and ID — it is safe to share with PathScribe support.

12. Demo Reset (Development / Staging Only)

Navigate to **Configuration** → **Demo Reset**. This tab is visible in development and staging environments only and will be removed before production deployment.

- **Reset my data only** — restores your hospital's cases and data without affecting other testers. Use between review sessions.
- **Full reset** — restores all data for all users across all hospitals. Use sparingly.
- After confirming, PathScribe reloads automatically with clean seed data.

13. Pre-Launch Implementation Checklist

All items below must be completed before a customer goes live on PathScribe by ForMedrix AI. Do not grant production access until every item is verified.

#	Requirement	Owner
1	Anthropic API key stored in backend secrets manager (never in VITE_ env vars)	Engineering
2	Business Associate Agreement (BAA) signed with Anthropic before any real PHI is processed	Legal
3	Anthropic training data opt-out confirmed for API account	Engineering
4	Production backend proxy built for all AI calls — Vite dev proxy is development only	Engineering
5	AMA CPT data licence procured (approx. \$3–10k/year) before CPT codes are displayed	Legal
6	CPT backend proxy endpoint built at /api/terminology/cpt	Engineering
7	UMLS licence obtained for full SNOMED CT (NLM Conditions API used in development only)	Legal
8	SNOMED CT production endpoint configured via UMLS API with backend key injection	Engineering
9	Non-US ICD-10 variant proxy configured if customer uses RCPATH, RCPA, or ICCR protocols	Engineering
10	Firestore encryption at rest confirmed enabled in GCP project	Engineering
11	Flag definitions seeded for customer jurisdiction and lab type	Implementation
12	Governing body configured for customer region (CAP / RCPATH / RCPA / ICCR)	Implementation
13	AI provider set to Claude Sonnet 4 (recommended for clinical accuracy)	Implementation
14	All Terminology Services health checks passing (Configuration → System → Terminology Services)	Implementation
15	PHI de-identification implemented before AI prompt submission with real patient data	Engineering
16	Penetration test completed by external vendor before go-live	Security

Quick Reference — Administration

Task	Navigation path
Add a staff user	Configuration → Staff → Add User
Deactivate a user	Configuration → Staff → [user] → Deactivate
Manage physician directory	Configuration → System → Physicians
Edit role permissions	Configuration → Staff → Roles → [role] → Permissions
Create a flag	Configuration → System → Flags → Add Flag
Configure computational flag	Configuration → System → Flags → [flag] → Computational section
Add a delegation type	Configuration → System → Delegation Types → + New Type
Open Template Builder	Configuration → Template Builder
Create a new Part	Template Builder → Part Library → New Part
Assemble a template	Template Builder → Templates → New Template
Publish a template	Template Assembly page → Publish
Change AI model	Configuration → AI Settings → Model
Set confidence threshold	Configuration → AI Settings → Confidence Threshold
Configure identifier formats	Configuration → System → Identifier Formats
Check terminology services	Configuration → System → Terminology Services
View action registry	Configuration → Action Registry
View audit log	Home → System Audit
System information	Nav bar initials badge → System Information
Generate support report	System Information → Copy to Clipboard
Reset demo data	Configuration → Demo Reset (development only)